

Policy Improvement in Infinite-Horizon MDP

Reinforcement Learning

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1 Setting and Notation

We consider an infinite-horizon discounted Markov decision process with state space $\mathcal{S}$, action sets $\mathcal{A}(s)$ for each $s \in \mathcal{S}$, reward space $\mathcal{R}$, transition probabilities $p(s' \mid s, a)$, reward distribution $p(r \mid s, a)$, and discount factor $0 \leq \gamma < 1$. We assume $\mathcal{S}$ and each $\mathcal{A}(s)$ are finite, and the one-step expected reward is bounded.

The expected immediate reward is defined by

$$r(s, a) := \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a] = \sum_{r \in \mathcal{R}} r p(r \mid s, a).$$

A stationary policy π assigns probabilities $\pi(a \mid s)$ over actions $a \in \mathcal{A}(s)$.

For a fixed stationary policy π , the state-value function and action-value function are

$$v_\pi(s) := \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s \right],$$

$$q_\pi(s, a) := \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s, A_t = a \right].$$

They satisfy the Bellman consistency equations

$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) q_\pi(s, a),$$

$$q_\pi(s, a) = r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s, a) v_\pi(s').$$

For deterministic π , this reduces to

$$v_\pi(s) = q_\pi(s, \pi(s)).$$

We also define the Bellman expectation operator

$$(T_\pi v)(s) := \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s, a) v(s') \right].$$

Then v_π is the unique fixed point of T_π :

$$v_\pi = T_\pi v_\pi.$$

We will also use the Bellman optimality operator

$$(Tv)(s) := \max_{a \in \mathcal{A}(s)} \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s, a) v(s') \right].$$

Its unique fixed point is the optimal value function v_* , and the associated optimal action-value function is

$$q_*(s, a) = r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s, a) v_*(s').$$

2 Goal of this Report

The purpose of this report is to prove the *Policy Improvement Theorem* in two ways, both in Sutton notation:

- (1) a complete componentwise proof in the spirit of Sutton, but written rigorously;
- (2) a complete operator proof, cleaner and closer to the Bellman-operator style used in discounted MDP analysis.

For a stationary policy π' we define

$$q_\pi(s, \pi') := \sum_{a \in \mathcal{A}(s)} \pi'(a | s) q_\pi(s, a).$$

In the deterministic case this reduces to

$$q_\pi(s, \pi') = q_\pi(s, \pi'(s)).$$

3 Mathematical Preliminaries

3.1 Monotonicity of the Bellman Expectation Operator

We first prove a basic monotonicity property of the Bellman expectation operator.

Lemma 3.1. *Let $u, w : \mathcal{S} \rightarrow \mathbb{R}$ satisfy*

$$u(s) \leq w(s) \quad \forall s \in \mathcal{S}.$$

Then

$$(T_\pi u)(s) \leq (T_\pi w)(s) \quad \forall s \in \mathcal{S}.$$

Proof. Fix $s \in \mathcal{S}$. Then

$$(T_\pi w)(s) - (T_\pi u)(s) = \sum_{a \in \mathcal{A}(s)} \pi(a | s) \left[\gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) (w(s') - u(s')) \right].$$

Since $\pi(a | s) \geq 0$, $p(s' | s, a) \geq 0$, $\gamma \geq 0$, and $w(s') - u(s') \geq 0$ for all s' , every term in the sum is nonnegative. Hence

$$(T_\pi w)(s) - (T_\pi u)(s) \geq 0.$$

Therefore

$$(T_\pi u)(s) \leq (T_\pi w)(s).$$

This holds for every $s \in \mathcal{S}$. □

4 The Policy Improvement Theorem

4.1 Statement

Let π and π' be two stationary policies. Because

$$q_\pi(s, a) = r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) v_\pi(s'),$$

we have

$$\begin{aligned}
q_\pi(s, \pi') &= \sum_{a \in \mathcal{A}(s)} \pi'(a | s) q_\pi(s, a) \\
&= \sum_{a \in \mathcal{A}(s)} \pi'(a | s) \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) v_\pi(s') \right] \\
&= (T_{\pi'} v_\pi)(s).
\end{aligned}$$

We now state the policy improvement theorem.

Theorem 4.1 (Policy Improvement Theorem). *Suppose π and π' are stationary policies such that*

$$q_\pi(s, \pi') \geq v_\pi(s) \quad \forall s \in \mathcal{S}.$$

Equivalently,

$$T_{\pi'} v_\pi \geq v_\pi \quad \text{componentwise.}$$

Then

$$v_{\pi'}(s) \geq v_\pi(s) \quad \forall s \in \mathcal{S}.$$

4.2 Proof I: Componentwise Recursive Expansion

Proof. Fix any state $s \in \mathcal{S}$. By assumption,

$$q_\pi(s, \pi') \geq v_\pi(s).$$

Using the definition of $q_\pi(s, \pi')$,

$$\sum_{a \in \mathcal{A}(s)} \pi'(a | s) q_\pi(s, a) \geq v_\pi(s).$$

Substitute the Bellman consistency formula for $q_\pi(s, a)$:

$$\sum_{a \in \mathcal{A}(s)} \pi'(a | s) \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) v_\pi(s') \right] \geq v_\pi(s).$$

Hence

$$v_\pi(s) \leq \sum_{a \in \mathcal{A}(s)} \pi'(a | s) \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) v_\pi(s') \right]. \quad (5.1)$$

Now apply the same inequality at every successor state s' . For each $s' \in \mathcal{S}$,

$$v_\pi(s') \leq \sum_{a' \in \mathcal{A}(s')} \pi'(a' | s') \left[r(s', a') + \gamma \sum_{s'' \in \mathcal{S}} p(s'' | s', a') v_\pi(s'') \right]. \quad (5.2)$$

Substitute (5.2) into (5.1). Since $\gamma \geq 0$, $\pi'(\cdot | s) \geq 0$, and $p(\cdot | s, a) \geq 0$, the inequality is preserved:

$$\begin{aligned}
v_\pi(s) &\leq \sum_{a \in \mathcal{A}(s)} \pi'(a | s) \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) \sum_{a' \in \mathcal{A}(s')} \pi'(a' | s') \right. \\
&\quad \cdot \left. \left(r(s', a') + \gamma \sum_{s'' \in \mathcal{S}} p(s'' | s', a') v_\pi(s'') \right) \right].
\end{aligned}$$

This is exactly the expected two-step return obtained by following π' for two steps and then using v_π as the tail value:

$$v_\pi(s) \leq \mathbb{E}_{\pi'}[R_{t+1} + \gamma R_{t+2} + \gamma^2 v_\pi(S_{t+2}) \mid S_t = s]. \quad (5.3)$$

Repeating this substitution argument n times yields

$$v_\pi(s) \leq \mathbb{E}_{\pi'} \left[\sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n v_\pi(S_{t+n}) \mid S_t = s \right]. \quad (5.4)$$

Now v_π is bounded because rewards are bounded and $0 \leq \gamma < 1$. Thus there exists $M < \infty$ such that

$$|v_\pi(s)| \leq M \quad \forall s \in \mathcal{S}.$$

Hence

$$|\gamma^n v_\pi(S_{t+n})| \leq \gamma^n M \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Taking $n \rightarrow \infty$ in (5.4), we obtain

$$v_\pi(s) \leq \mathbb{E}_{\pi'} \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s \right].$$

The right-hand side is precisely $v_{\pi'}(s)$. Therefore

$$v_\pi(s) \leq v_{\pi'}(s).$$

Since $s \in \mathcal{S}$ was arbitrary,

$$v_{\pi'}(s) \geq v_\pi(s) \quad \forall s \in \mathcal{S}.$$

This proves the theorem. □

4.2.1 Detailed Term Expansion via the Law of Total Expectation

We now expand the first few terms in (5.3) carefully using the law of total expectation and the Markov property. This makes the recursive substitution argument completely explicit.

Step 1: Expansion of $\mathbb{E}_{\pi'}[R_{t+1} \mid S_t = s]$. Since under π' the action A_t is drawn according to $\pi'(\cdot \mid s)$, we have

$$\begin{aligned} \mathbb{E}_{\pi'}[R_{t+1} \mid S_t = s] &= \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s) \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a] \\ &= \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s) r(s, a). \end{aligned}$$

Step 2: Expansion of $\mathbb{E}_{\pi'}[R_{t+2} \mid S_t = s]$. Again conditioning first on the action A_t ,

$$\mathbb{E}_{\pi'}[R_{t+2} \mid S_t = s] = \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s) \mathbb{E}[R_{t+2} \mid S_t = s, A_t = a].$$

Now condition on the next state S_{t+1} :

$$\mathbb{E}[R_{t+2} \mid S_t = s, A_t = a] = \sum_{s' \in \mathcal{S}} p(s' \mid s, a) \mathbb{E}[R_{t+2} \mid S_t = s, A_t = a, S_{t+1} = s'].$$

Next condition on A_{t+1} chosen according to $\pi'(\cdot \mid s')$:

$$\mathbb{E}[R_{t+2} \mid S_t = s, A_t = a, S_{t+1} = s'] = \sum_{a' \in \mathcal{A}(s')} \pi'(a' \mid s') \mathbb{E}[R_{t+2} \mid S_t = s, A_t = a, S_{t+1} = s', A_{t+1} = a'].$$

By the Markov property,

$$R_{t+2} \perp (S_t, A_t) \mid (S_{t+1}, A_{t+1}),$$

so

$$\mathbb{E}[R_{t+2} \mid S_t = s, A_t = a, S_{t+1} = s', A_{t+1} = a'] = \mathbb{E}[R_{t+2} \mid S_{t+1} = s', A_{t+1} = a'] = r(s', a').$$

Therefore,

$$\mathbb{E}_{\pi'}[R_{t+2} \mid S_t = s] = \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s) \sum_{s' \in \mathcal{S}} p(s' \mid s, a) \sum_{a' \in \mathcal{A}(s')} \pi'(a' \mid s') r(s', a').$$

Step 3: Expansion of $\mathbb{E}_{\pi'}[v_\pi(S_{t+2}) \mid S_t = s]$. We expand similarly:

$$\mathbb{E}_{\pi'}[v_\pi(S_{t+2}) \mid S_t = s] = \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s) \mathbb{E}[v_\pi(S_{t+2}) \mid S_t = s, A_t = a].$$

Condition on S_{t+1} :

$$\mathbb{E}[v_\pi(S_{t+2}) \mid S_t = s, A_t = a] = \sum_{s' \in \mathcal{S}} p(s' \mid s, a) \mathbb{E}[v_\pi(S_{t+2}) \mid S_t = s, A_t = a, S_{t+1} = s'].$$

Condition next on A_{t+1} :

$$\mathbb{E}[v_\pi(S_{t+2}) \mid S_t = s, A_t = a, S_{t+1} = s'] = \sum_{a' \in \mathcal{A}(s')} \pi'(a' \mid s') \mathbb{E}[v_\pi(S_{t+2}) \mid S_t = s, A_t = a, S_{t+1} = s', A_{t+1} = a'].$$

Now by the Markov property, S_{t+2} depends on the past only through (S_{t+1}, A_{t+1}) , so

$$\mathbb{E}[v_\pi(S_{t+2}) \mid S_t = s, A_t = a, S_{t+1} = s', A_{t+1} = a'] = \mathbb{E}[v_\pi(S_{t+2}) \mid S_{t+1} = s', A_{t+1} = a'].$$

Finally, condition on S_{t+2} itself:

$$\mathbb{E}[v_\pi(S_{t+2}) \mid S_{t+1} = s', A_{t+1} = a'] = \sum_{s'' \in \mathcal{S}} p(s'' \mid s', a') v_\pi(s'').$$

Therefore,

$$\mathbb{E}_{\pi'}[v_\pi(S_{t+2}) \mid S_t = s] = \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s) \sum_{s' \in \mathcal{S}} p(s' \mid s, a) \sum_{a' \in \mathcal{A}(s')} \pi'(a' \mid s') \sum_{s'' \in \mathcal{S}} p(s'' \mid s', a') v_\pi(s'').$$

Multiplying by γ^2 , we get

$$\mathbb{E}_{\pi'}[\gamma^2 v_\pi(S_{t+2}) \mid S_t = s] = \gamma^2 \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s) \sum_{s' \in \mathcal{S}} p(s' \mid s, a) \sum_{a' \in \mathcal{A}(s')} \pi'(a' \mid s') \sum_{s'' \in \mathcal{S}} p(s'' \mid s', a') v_\pi(s'').$$

Putting the first few terms together. Combining the three expansions above,

$$\begin{aligned}
& \mathbb{E}_{\pi'} [R_{t+1} + \gamma R_{t+2} + \gamma^2 v_{\pi}(S_{t+2}) \mid S_t = s] \\
&= \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s) r(s, a) \\
&\quad + \gamma \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s) \sum_{s' \in \mathcal{S}} p(s' \mid s, a) \sum_{a' \in \mathcal{A}(s')} \pi'(a' \mid s') r(s', a') \\
&\quad + \gamma^2 \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s) \sum_{s' \in \mathcal{S}} p(s' \mid s, a) \sum_{a' \in \mathcal{A}(s')} \pi'(a' \mid s') \sum_{s'' \in \mathcal{S}} p(s'' \mid s', a') v_{\pi}(s'').
\end{aligned}$$

This is exactly the explicit componentwise version of the two-step expansion used in (5.3).

4.2.2 Intuition Behind the Recursive Expansion

The recursive expansion proof has a very simple meaning.

We start from the inequality

$$q_{\pi}(s, \pi') \geq v_{\pi}(s),$$

which says the following: if we stand at state s , take actions according to π' *for one step*, and then switch back to following π , then this one-step deviation is at least as good as following π immediately.

The proof then repeats the same argument at the next state, then at the next state after that, and so on. After two expansions, we are following π' for two steps and then π afterward. After three expansions, we are following π' for three steps and then π afterward. After n expansions, we are following π' for n steps and then π afterward.

So the proof constructs the chain

$$\text{follow } \pi \leq \text{follow } \pi' \text{ for 1 step, then } \pi \leq \text{follow } \pi' \text{ for 2 steps, then } \pi \leq \dots$$

Each new expression is no worse than the previous one.

Finally, as $n \rightarrow \infty$, the “follow π' for n steps, then π ” strategy becomes exactly “follow π' forever.” The discount factor $\gamma < 1$ ensures that the leftover tail term

$$\gamma^n v_{\pi}(S_{t+n})$$

vanishes in the limit. Therefore the return under π' is at least as large as the return under π .

In short, the proof means:

If it is never worse to replace one step of π by one step of π' , then it is never worse to replace two steps, or three steps, or eventually all steps. Hence following π' forever is at least as good as following π forever.

This is the basic idea of policy improvement in its most direct componentwise form.

4.3 Proof II: Bellman Operator Method

Proof. Assume

$$q_\pi(s, \pi') \geq v_\pi(s) \quad \forall s \in \mathcal{S}.$$

As observed above, this is equivalent to

$$(T_{\pi'} v_\pi)(s) \geq v_\pi(s) \quad \forall s \in \mathcal{S},$$

that is,

$$T_{\pi'} v_\pi \geq v_\pi \quad \text{componentwise.} \quad (7.1)$$

Apply the monotonicity lemma to $T_{\pi'}$. From (7.1),

$$T_{\pi'} v_\pi \geq v_\pi.$$

Applying $T_{\pi'}$ to both sides and using monotonicity gives

$$T_{\pi'}^2 v_\pi \geq T_{\pi'} v_\pi.$$

Applying $T_{\pi'}$ again,

$$T_{\pi'}^3 v_\pi \geq T_{\pi'}^2 v_\pi.$$

Continuing inductively, we obtain the monotone chain

$$v_\pi \leq T_{\pi'} v_\pi \leq T_{\pi'}^2 v_\pi \leq T_{\pi'}^3 v_\pi \leq \dots \quad (7.2)$$

Now $T_{\pi'}$ is a γ -contraction in the sup norm, and $v_{\pi'}$ is its unique fixed point. Therefore, for any initial bounded function v ,

$$T_{\pi'}^n v \rightarrow v_{\pi'} \quad \text{as } n \rightarrow \infty.$$

In particular, taking $v = v_\pi$,

$$T_{\pi'}^n v_\pi \rightarrow v_{\pi'}. \quad (7.3)$$

Taking limits in the monotone chain (7.2) and using (7.3), we conclude

$$v_\pi \leq v_{\pi'} \quad \text{componentwise.}$$

Thus

$$v_{\pi'}(s) \geq v_\pi(s) \quad \forall s \in \mathcal{S}.$$

This proves the theorem. □

5 Greedy Policy Corollary

5.1 Componentwise Form

A particularly important special case is when π' is greedy with respect to q_π , meaning that for each $s \in \mathcal{S}$,

$$\pi'(\cdot \mid s) \text{ places all probability on actions in } \arg \max_{a \in \mathcal{A}(s)} q_\pi(s, a).$$

Then automatically

$$q_\pi(s, \pi') \geq v_\pi(s) \quad \forall s \in \mathcal{S},$$

because $v_\pi(s) = \sum_a \pi(a \mid s) q_\pi(s, a)$ is a weighted average of the values $q_\pi(s, a)$, while $q_\pi(s, \pi')$ puts all its mass on maximizing actions. Thus the policy improvement theorem yields

$$v_{\pi'}(s) \geq v_\pi(s) \quad \forall s \in \mathcal{S}.$$

5.2 Operator Form

Suppose π' is greedy with respect to v_π , meaning

$$\pi'(s) \in \arg \max_{a \in \mathcal{A}(s)} \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) v_\pi(s') \right] \quad \forall s \in \mathcal{S}.$$

Then by definition of T ,

$$(T_{\pi'} v_\pi)(s) = (T v_\pi)(s) \quad \forall s \in \mathcal{S}.$$

Also, since $v_\pi = T_\pi v_\pi$,

$$\begin{aligned} (T v_\pi)(s) &= \max_{a \in \mathcal{A}(s)} \left[r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) v_\pi(s') \right] \\ &\geq (T_\pi v_\pi)(s) = v_\pi(s). \end{aligned}$$

Hence

$$T_{\pi'} v_\pi \geq v_\pi.$$

Applying the policy improvement theorem,

$$v_{\pi'} \geq v_\pi.$$

This is the operator form of policy improvement.